Calcurion Physics Worksheets

A-Level Physics

Topic: Electricity

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Define electric current. **[1]**

2. State the equation $I = nAvq$ and define each symbol. **[3]**

3. A copper wire has $n = 8.5 \times 10^{28}$ m^{-3}, $A = 1.0 \times 10^{-6}$ m^2, $I = 2.0$ A. Calculate drift speed v. ($e = 1.6 \times 10^{-19}$ C) **[3]**

4. Define potential difference. **[1]**

5. State Ohm's law and the conditions under which it applies. **[2]**

6. A resistor has $V = 6.0$ V and $I = 0.25$ A. Calculate resistance and power dissipated. **[3]**

7. Define resistivity and state its unit. **[2]**

8. A wire of length 2.0 m, area 0.50 mm^2 and resistance 0.80 Ω has resistivity? **[3]**

9. State Kirchhoff's first law. **[1]**

10. State Kirchhoff's second law. **[2]**

11. Two resistors 4.0 Ω and 6.0 Ω in series with a 12 V supply. Calculate current and p.d. across each. **[3]**

12. The same two resistors in parallel with 12 V. Calculate total resistance and total current. **[3]**

13. Define e.m.f. of a cell. **[1]**

14. A cell of e.m.f. 1.5 V and internal resistance 0.50 Ω is connected to a 2.5 Ω resistor. Calculate terminal p.d. **[3]**

15. Sketch the I–V characteristic of a semiconductor diode. **[2]**

Section B – Hard (Q16–25)

16. Derive $P = I^2R$ from $P = IV$ and $V = IR$. **[2]**

17. A cable of resistivity 1.7×10^{-8} Ω m, length 50 m and cross-section 2.5 mm^2 carries 10 A. Calculate power loss in the cable. **[4]**

18. A cell of e.m.f. $\mathcal{E}$ and internal resistance r has terminal p.d. V when current I flows. Show that $V = \mathcal{E} - Ir$. **[3]**

19. In a potential divider, two resistors $R_1 = 2$ kΩ and $R_2 = 3$ kΩ share 10 V. Calculate the output across R_2. **[3]**

20. Explain why the resistance of a metallic conductor increases with temperature. **[3]**

21. A circuit has three resistors: 2 Ω and 3 Ω in parallel, then in series with 4 Ω, supplied by 12 V. Calculate total resistance and current from the supply. **[4]**

22. Calculate the charge that flows when a current of 150 mA passes for 2.0 minutes. **[2]**

23. A 12 V battery of internal resistance 1.0 Ω is connected to two 5.0 Ω resistors in parallel. Calculate the current through the battery. **[4]**

24. Define the kilowatt-hour and calculate the cost of running a 2.5 kW heater for 3 hours at 15 p per kWh. **[3]**

25. Describe an experiment to determine the resistivity of a wire. **[4]**

End of Worksheet • *Show all working clearly.*